

FULL PAPER

DEVELOPMENT OF AN EXPLAINABLE AND
UNCERTAINTY-AWARE AI SYSTEM FOR CERVICAL
CELL ABNORMALITY SCREENING AND HPV-
RELATED MORPHOLOGY RISK ASSESSMENT
FROM THINPREP-STYLE IMAGES

INSERT
INSTITUTION LOGO

COMPILED BY

AUTHOR 1: _____
AUTHOR 2: _____
AUTHOR 3: _____
AUTHOR 4: _____
AUTHOR 5: _____
AUTHOR 6: _____
INSTITUTION _____
CATEGORY TECHNOLOGY
YEAR 2026

TABLE OF CONTENTS

This table follows the final Word/PDF pagination of the polished submission version.

Abstract	3
Chapter 1 - Introduction	4
Chapter 2 - Literature Review	5
Chapter 3 - Research Method	6
Chapter 4 - Results and Discussion	8
Chapter 5 - Conclusion	11
References	12

Submission format *English; A4; Arial 12 pt; single spacing; justified; margins 4/3/3/3 cm; PDF and Microsoft Word.*

ABSTRACT

Cervical cancer is largely preventable when screening, interpretation, communication, and follow-up operate as a continuous pathway. This study developed CerviCo-Pilot, an explainable and uncertainty-aware research prototype for clinician-in-the-loop cervical cytology screening support. The system separates a four-class cytology-grade output (NILM, LSIL, HSIL, and SCC) from an independent koilocytosis-morphology endpoint, and adds a binary normal/abnormal safety-triage view, Grad-CAM visual review support, Monte Carlo Dropout uncertainty, and a report lock that requires clinician sign-off before patient-facing communication. An EfficientNet-B0 grade model was trained and evaluated using 917 real images from the public Herlev cervical cytology dataset; segmentation masks were excluded from performance evidence. On the held-out test set (n=137), the legacy five-output evaluation had accuracy 0.6934, macro F1 0.5545, macro AUROC was 0.7311, and quadratic weighted kappa was 0.687. The binary safety-triage view achieved sensitivity 1.000, specificity 0.7222, AUROC 0.964, AUPRC 0.9856, and a confusion matrix of TP=101, TN=26, FP=10, and FN=0. Five-fold cross-validation produced binary sensitivity of 0.9867 +/- 0.0086 and AUROC of 0.9435 +/- 0.0448. Post-hoc temperature scaling improved held-out calibration metrics without materially changing discrimination. For the separate KOIL endpoint, EfficientNet-B0 was trained on all 4,049 official SIPaKMeD cropped cells using source-cluster-disjoint splits. On the locked test set (641 cells; 133 KOIL-positive), sensitivity was 0.9624, specificity 0.9764, AUROC 0.9912, AUPRC 0.9810, and ECE 0.0134; cluster-bootstrap 95% confidence intervals were 0.9167-0.9921 for sensitivity and 0.9583-0.9916 for specificity. The prototype demonstrates a complete review workflow rather than an autonomous diagnostic device. Herlev and SIPaKMeD are public conventional cytology domains; the system does not replace molecular HPV testing, and Thai ThinPrep validation, paired HPV endpoints, reader studies, and prospective evaluation remain necessary before clinical use.

Keywords: *cervical cytology, explainable artificial intelligence, uncertainty-aware triage, Grad-CAM, clinician-in-the-loop, ThinPrep, koilocytosis*

Evidence boundary *Grade and triage evidence comes from Herlev; KOIL morphology evidence comes from SIPaKMeD conventional Pap-smear crops. Neither is Thai ThinPrep validation or molecular HPV testing.*

CHAPTER 1

INTRODUCTION

1.1 Research Background

Cervical cancer remains an important public-health problem, yet it can be prevented or treated early when vaccination, screening, diagnosis, and follow-up are connected [1,2]. Cytology-based screening, including Pap and liquid-based preparations such as ThinPrep, examines cellular morphology, while HPV molecular testing detects viral material; the two modalities must not be described as interchangeable [3,4].

The practical gap is not limited to image classification. Screening programs also depend on timely review, understandable communication, and reliable referral. In settings with limited cytology expertise, a compact offline decision-support tool may help prioritize abnormal images, expose the image regions that influenced a prediction, and return uncertain cases to human review.

1.2 Problem Statement

Many image classifiers provide only a label and confidence score. Such output is insufficient for a safety-sensitive workflow because confidence may be miscalibrated, explanations may be over-trusted, and patient-facing communication may be released without appropriate review. A useful prototype should combine grading, triage, explainability, uncertainty, and explicit clinician control.

1.3 Research Objectives

- Develop a four-class Bethesda-style cervical cytology grading model.
- Develop a separate koilocytosis-morphology endpoint without treating KOIL as a Bethesda grade.
- Add a high-sensitivity binary safety-triage view without replacing the grade output.
- Provide Grad-CAM review support and uncertainty-aware abstention.
- Demonstrate clinician sign-off, report locking, and offline operation.
- Define a validation pathway for Thai ThinPrep images and paired HPV endpoints.

1.4 Significance

The project contributes an integrated medical-AI workflow rather than an isolated classifier. It emphasizes reviewability, conservative communication, reproducible metrics, and explicit boundaries between current evidence and future clinical validation.

CHAPTER 2

LITERATURE REVIEW

2.1 Cervical Cytology and the Bethesda Framework

The Bethesda System standardizes cervical cytology terminology and distinguishes negative findings from low-grade, high-grade, and malignant squamous abnormalities [4]. Preserving multiple classes supports clinical communication, whereas a secondary binary view is useful for screening-safety analysis.

2.2 Image Classification and Explainability

The Herlev benchmark contains 917 single-cell cervical cytology images [5]. SIPaKMeD provides 4,049 cropped cells from 966 source cluster images across five morphology classes, including 825 koilocytotic cells [14]. EfficientNet provides an efficient scaling strategy for offline image models [6]. Grad-CAM localizes regions associated with a class score [7], but it remains a review aid rather than causal proof.

2.3 Uncertainty, Calibration, and Governance

Monte Carlo Dropout provides an approximate uncertainty signal [8], while temperature scaling can improve probability calibration without changing class ordering [9]. Medical-AI guidance emphasizes representative data, human-AI interaction, lifecycle monitoring, and transparent reporting [10-12]. Dataset shift remains a major risk when moving from a public benchmark to a different population, preparation method, microscope, or camera [13].

Table 1. Design principles derived from the literature.

Principle	System implementation	Boundary
Structured cytology	Four-class Bethesda-style grade	Not a final cytology report
KOIL morphology	Independent one-vs-rest endpoint	Not HPV DNA/RNA status
Screening safety	Binary normal/abnormal triage	Not the sole product output
Explainability	Grad-CAM heatmap	Review aid, not causal proof
Uncertainty	MC Dropout and abstention	Requires external validation
Governance	Clinician sign-off and report lock	Demo workflow only

CHAPTER 3

RESEARCH METHOD

3.1 Study Design and Dataset

This retrospective model-development and prototype-integration study used two public datasets. Grade and binary-triage evidence used 917 real Herlev images [5], with segmentation masks and synthetic experiments excluded. The independent KOIL endpoint used all 4,049 official SIPaKMeD cropped cells from 966 source clusters [14].

Table 2. Dataset and model card summary.

Item	Specification
Grade/triage dataset	Herlev: 917 real images; masks excluded
KOIL dataset	SIPaKMeD: 4,049 cells from 966 source clusters
Architecture	EfficientNet-B0 with ImageNet transfer learning
Grade output	NILM, LSIL, HSIL, SCC
KOIL output	Independent koilocytosis morphology probability
Evidence status	Internal public-dataset validation; no Thai ThinPrep evidence

3.2 End-to-End Workflow

The workflow accepts a cytology image, returns four grade probabilities, binary triage, and an independent KOIL morphology assessment. Grade and KOIL use class-specific Grad-CAM review. Uncertainty and image-quality gates route unsafe cases to clinician confirmation, correction, or rejection; patient-facing output remains locked until all gates pass.



Safety: high uncertainty blocks patient report; AI does not replace HPV DNA/RNA testing or clinician review.

Figure 1. CerviCo-Pilot clinician-in-the-loop workflow.

CHAPTER 3

RESEARCH METHOD (CONTINUED)

3.3 Model Development and Evaluation

Images were resized for EfficientNet-B0 input and processed through canonical pipelines. Herlev grade/triage metrics were reproduced from models/best_cervical.pt. For SIPaKMeD, source clusters were split before training (676/145/145 clusters for train/validation/test), preventing cropped cells from the same cluster crossing splits. Model selection, temperature scaling, and a sensitivity-constrained KOIL threshold were fixed on validation data before one-time test evaluation.

Table 3. Reproducible evaluation specification.

Component	Specification	Purpose
Held-out evaluation	n=137; TTA enabled	Primary test evidence
KOIL locked test	641 cells; 145 clusters	Independent endpoint evidence
Bootstrap	2,000 percentile resamples	Confidence intervals
Cross-validation	Five folds; mean +/- SD	Split robustness
Binary threshold	0.50	Normal/abnormal triage
Calibration	Temperature=0.794103	Post-hoc probability scaling
KOIL calibration	Temperature=0.838126	Validation-fit; test locked
Explainability	Grad-CAM	Visual review support
Uncertainty	MC Dropout	Abstention signal

3.4 Metrics and Safety Policy

Legacy grade evaluation included accuracy, macro F1, macro AUROC, per-class recall, and quadratic weighted kappa. Binary triage and KOIL one-vs-rest evaluation included sensitivity, specificity, AUROC, AUPRC, precision, F1, and confusion counts. KOIL confidence intervals used 2,000 source-cluster bootstrap resamples. Calibration used ECE and Brier score. High uncertainty triggers abstention and blocks patient-report release.

3.5 Prototype System

The React/FastAPI prototype supports sample review, image upload, separate grade and KOIL probabilities, class-specific Grad-CAM, confirm/edit/reject actions, report export, and an audit demonstration. It is intended for supervised research demonstration and does not satisfy regulated clinical deployment requirements.

Do-not-use boundary *No autonomous diagnosis, molecular HPV status, unsupervised patient release, or Thai-domain performance claim.*

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Primary Quantitative Results

Table 4. Held-out and cross-validation results.

Measure	Result	Interpretation
Legacy five-output accuracy	0.6934	KOIL unsupported in Herlev
Legacy macro F1 / AUROC	0.5545 / 0.7311	Includes unsupported KOIL output
Quadratic weighted kappa	0.687	Severity agreement
Binary sensitivity / specificity	1.000 / 0.7222	FN=0; some over-referral
Binary AUROC / AUPRC	0.964 / 0.9856	Strong Herlev triage
Five-fold sensitivity	0.9867 +/- 0.0086	High across folds
Five-fold AUROC	0.9435 +/- 0.0448	Within-domain robustness

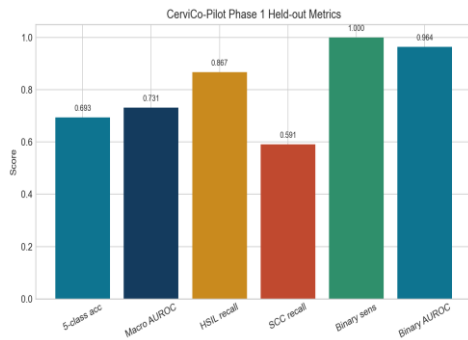


Figure 2. Held-out headline metrics.

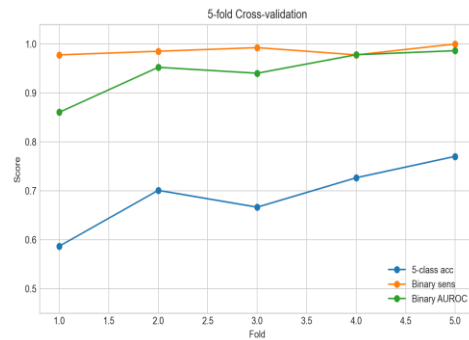


Figure 3. Five-fold cross-validation.

The historical checkpoint used a five-output head, but Herlev contained no true KOIL support; the deployed interface therefore exposes only four grade classes and obtains KOIL morphology from the separate SIPaKMeD model. Binary triage achieved high sensitivity and remains a safety view rather than a replacement for detailed grade review.

CHAPTER 4

RESULTS AND DISCUSSION (CONTINUED)

4.2 Grade Error Analysis

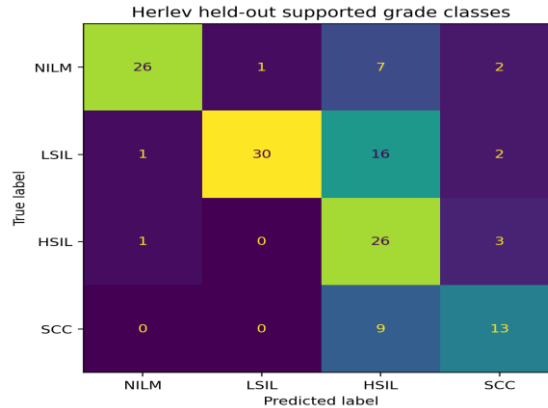


Figure 4. Herlev confusion matrix for the four supported grade classes.

Table 5. Herlev held-out per-class recall.

Class	Support	Recall	Interpretation
NILM	36	0.7222	False positives reduce specificity
LSIL	49	0.6122	Overlap with higher-grade morphology
HSIL	30	0.8667	Strongest high-grade recall
SCC	22	0.5909	Requires improvement

4.3 Independent KOIL Morphology Endpoint

Table 6. Locked SIPaKMeD KOIL one-vs-rest test results.

Measure	Result	Cluster-bootstrap 95% CI
Support	133 positive / 508 negative	145 source clusters
Sensitivity / specificity	0.9624 / 0.9764	0.9167-0.9921 / 0.9583-0.9916
Precision / F1	0.9143 / 0.9377	0.8291-0.9702 / 0.8811-0.9706
AUROC / AUPRC	0.9912 / 0.9810	0.9786-0.9984 / 0.9506-0.9951

At the validation-locked threshold of 0.3367, the model produced 128 TP, 496 TN, 12 FP, and 5 FN cells. False positives were limited to dyskeratotic (n=9) and metaplastic (n=3) morphology; each false negative came from a different source cluster. These errors require expert review and do not represent HPV test errors.

CHAPTER 4

RESULTS AND DISCUSSION (CONTINUED)

4.4 Calibration, Explainability, and System Interpretation

Table 7. Held-out calibration evidence.

Endpoint / measure	Before	After
Multiclass ECE	0.066853	0.038670
Binary ECE	0.090679	0.073787
Binary Brier score	0.071015	0.066994
KOIL ECE	Not reported	0.013385
KOIL Brier score	Not reported	0.019025

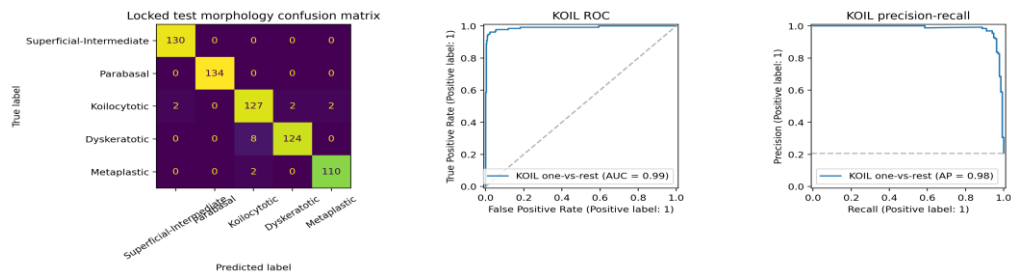


Figure 5. SIPaKMeD morphology confusion matrix and locked-test KOIL ROC/precision-recall curves.

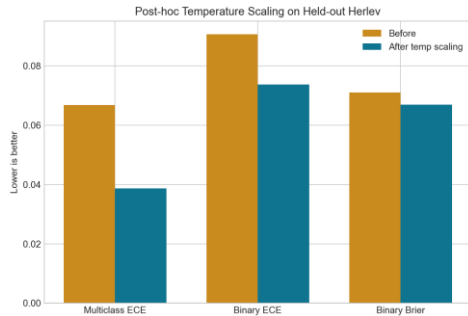


Figure 6. Herlev calibration.

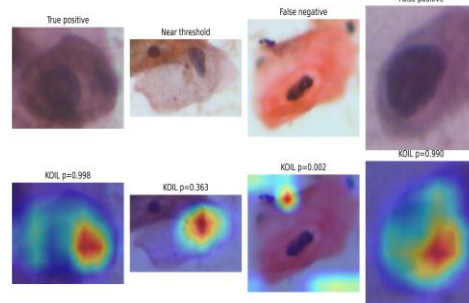


Figure 7. KOIL-specific Grad-CAM audit cases.

Temperature scaling improved calibration without changing class ordering [9]. KOIL Grad-CAM was generated for the independent endpoint and is presented as attention evidence, not segmentation or causal proof. Calibration and attribution remain domain-dependent and cannot be extended to Thai ThinPrep images without external evidence.

The prototype integrates evaluation and review: explanations can be challenged, uncertainty can trigger abstention, and patient communication is gated by clinician action [10-12]. Dataset shift remains the principal translation risk [13].

Required next evidence Thai ThinPrep external validation, paired HPV endpoints, pathologist error review, an unaided-versus-AI-assisted reader study, and a prospective workflow pilot.

CHAPTER 5

CONCLUSION

5.1 Conclusion

CerviCo-Pilot demonstrates an end-to-end, explainable, and uncertainty-aware cervical cytology screening-support prototype. On Herlev, binary triage achieved held-out sensitivity 1.000 and AUROC 0.964, while the legacy five-output grade evaluation achieved accuracy 0.6934. On the separate locked SIPaKMeD test set, KOIL morphology sensitivity was 0.9624 and AUROC 0.9912. These within-dataset results support continued research but do not justify autonomous diagnosis or deployment.

The main contribution is the system design: detailed cytology output is preserved, a binary view emphasizes screening safety, Grad-CAM supports visual review, uncertainty enables abstention, and clinician sign-off controls patient-facing communication. The system can operate locally and presents its limitations alongside its predictions.

5.2 Limitations

- Evidence is restricted to two public single-cell conventional cytology datasets.
- No Thai ThinPrep/LBC external validation or paired molecular HPV endpoint is available.
- KOIL morphology is internally validated on SIPaKMeD only; SCC grade recall remains limited.
- Grad-CAM, uncertainty, and report wording have not been evaluated in a clinical reader study.
- The web audit and report export remain research-demo components.

5.3 Future Work

The next phase will collect de-identified Thai ThinPrep/LBC images under ethics and data-governance approval, lock an external test set, add paired HPV labels, perform pathologist-reviewed error analysis, and conduct a reader study. A prospective pilot should measure review time, notification time, follow-up completion, and automation bias.

5.4 Research Integrity Statement

All performance values were copied from canonical evaluation files for the shipped Herlev grade/triage and SIPaKMeD KOIL checkpoints. External sources support background and methods only; they do not create CerviCo-Pilot performance evidence.

REFERENCES

- [1] World Health Organization, Cervical cancer, WHO Fact Sheet, 2026. Available: <https://www.who.int/news-room/fact-sheets/detail/cervical-cancer>.
- [2] World Health Organization, Global Strategy to Accelerate the Elimination of Cervical Cancer as a Public Health Problem, WHO, 2020.
- [3] Centers for Disease Control and Prevention, Screening for Cervical Cancer, CDC, 2025. Available: <https://www.cdc.gov/cervical-cancer/screening/>.
- [4] R. Nayar and D. C. Wilbur, Eds., The Bethesda System for Reporting Cervical Cytology, 3rd ed., Springer, 2015.
- [5] J. Jantzen, J. Norup, G. Dounias, and B. Bjerregaard, Pap-smear benchmark data for pattern classification, NiSIS 2005, 2005.
- [6] M. Tan and Q. V. Le, EfficientNet: Rethinking model scaling for convolutional neural networks, Proc. ICML, 2019, pp. 6105-6114.
- [7] R. R. Selvaraju et al., Grad-CAM: Visual explanations from deep networks via gradient-based localization, Proc. ICCV, 2017, pp. 618-626.
- [8] Y. Gal and Z. Ghahramani, Dropout as a Bayesian approximation: Representing model uncertainty in deep learning, Proc. ICML, 2016, pp. 1050-1059.
- [9] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, On calibration of modern neural networks, Proc. ICML, 2017, pp. 1321-1330.
- [10] U.S. FDA, Health Canada, and MHRA, Good Machine Learning Practice for Medical Device Development: Guiding Principles, 2021.
- [11] X. Liu et al., Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension, Nature Medicine, vol. 26, pp. 1364-1374, 2020.
- [12] B. Vasey et al., Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI, Nature Medicine, vol. 28, pp. 924-933, 2022.
- [13] S. G. Finlayson et al., The clinician and dataset shift in artificial intelligence, New England Journal of Medicine, vol. 385, pp. 283-286, 2021.
- [14] M. E. Plissiti et al., SiPaKMeD: A new dataset for feature and image based classification of normal and pathological cervical cells in Pap smear images, Proc. IEEE ICIP, 2018, pp. 3144-3148.